

Name: AAVISHKAR SINGH AAVISHKAR SINGH | Reg. No: RA2511003011024

Unit “ 4 : Session “ 2 : SLO “ 1 | Odd Man Out “ ALU Operations

■ I. Odd Man Out

1. Group ADD, SUBTRACT, MULTIPLY, FETCH ‘ FETCH “ it is an instruction-cycle step, not an ALU arithmetic operation.
2. Group AND, OR, NOT, LOAD ‘ LOAD “ it is a data-transfer operation, not a logical operation.
3. Group SHIFT LEFT, SHIFT RIGHT, ROTATE, COMPARE ‘ COMPARE “ it only sets condition flags; the others physically move bits.
4. Group INCREMENT, DECREMENT, DIVIDE, NEGATE ‘ DIVIDE “ it is a binary (two-operand) operation; the rest are unary (single-operand) operations.
5. Group XOR, XNOR, MOV, NAND ‘ MOV “ it transfers data; the others are Boolean logic operations.
6. Group CARRY FLAG, ZERO FLAG, OVERFLOW FLAG, CLOCK SIGNAL ‘ CLOCK SIGNAL “ it synchronizes the CPU; the others are status/condition flags.
7. Group ARITHMETIC, LOGIC, BRANCH, BITWISE ‘ BRANCH “ it is a control-flow operation; the others are ALU data-processing categories.
8. Group REGISTER, ADDER, COMPARATOR, MULTIPLEXER ‘ REGISTER “ it is a storage (sequential) element; the others are combinational circuits.
9. Group BINARY OPERATION, UNARY OPERATION, DATA TRANSFER, FLOATING POINT ‘ FLOATING POINT “ it is a data format/type, not an operation class.
10. Group BOOTH'S ALGORITHM, RESTORING DIVISION, CACHE HIT, WALLACE TREE ‘ CACHE HIT “ it is a memory-hierarchy event; the others are arithmetic hardware algorithms.